Day 5 of 15 · Student Activity Sheets

Day 5 Activities — Aggregates, JOINs and Export to Excel

TESDA Data Analytics Level III · Batch DAL3-2026

Name: _____ Partner: _____ Date: _____

Folder you need: `Day05/assets/Day05_SQL_Lab/` (contains `aling_nena.db` and `Day05_Queries.sql`)

Before you begin:

1. Copy `Day05_SQL_Lab` into `DAL3/Day05/`. Rename the script to `Day05_<Surname1>_<Surname2>.sql` (pair work from 5.2 onward).
2. Open the folder in VS Code (**Ctrl + K, Ctrl + O**), open your script, type each answer under its question.
3. Run: **select** the query → **Ctrl + E, Ctrl + E**. Write the answer or row count here. Save often (**Ctrl + S**).
4. **Pairs:** the **driver** types, the **navigator** reads and checks. **Switch every 2 queries** and fill in the pair log on the last page.

Activity	Title	Mode	Time	Points
5.0	Warm-up: Join by Hand	Individual	15 min	10
5.1	Aggregates and Aliases (Q1–Q7)	Individual	35 min	20
5.2	INNER JOIN, Two Tables (Q8–Q10)	Pair	30 min	20
5.3	The Joined Dataset, Three Tables (Q11)	Pair	20 min	10
5.4	LEFT JOIN: What Is Missing? (Q12–Q14)	Pair	25 min	15
5.5	Export to CSV, Import into Excel, Reconcile	Pair	35 min	20
	Pair log	Pair	5 min	5
	Total		**≈ 2 hrs 45 min**	**100**

Activity 5.0 — Warm-up: Join by Hand (10 pts)

No computer. Match each sale to the product list, like a JOIN does.

sales (part)

sale_id	product_id	quantity
TXN-90001	205	3
TXN-90002	201	1
TXN-90003	203	2

sale_id	product_id	quantity
TXN-90004	205	1
TXN-90005	209	4

products (part)

product_id	product_name	unit_price
201	Coca-Cola 1.5L	78
203	Pandesal (10 pcs)	50
205	Ligo Sardines 155g	25
207	Eggs (1 dozen)	96

a) Fill in the **INNER JOIN** result:

sale_id	product_name	quantity	amount (qty × price)

b) Which sale has **no matching product**? _____ What happens to it in an INNER JOIN? _____
c) Which product was **never sold** in this list? _____ Which kind of JOIN would show it? _____

Activity 5.1 — Aggregates and Aliases (20 pts)

Use the alias name given in each question.

Q	Task	Your answer
Q1	How many sales (rows) are there? Alias `total_sales`.	
Q2	Total revenue, the sum of `total_amount`. Alias `total_revenue`.	₱
Q3	Average sale amount, **rounded to 2 decimals**. Alias `avg_sale`.	₱
Q4	Revenue from **GCash** sales. Alias `gcash_revenue`.	₱
Q5	Cheapest and most expensive **unit price** in `products`. Aliases `lowest_price`, `highest_price`.	₱ ___ / ₱ ___
Q6	How many **different** suki customers bought something? Alias `buying_customers`.	
Q7	How many sales were **walk-in** sales (no customer ID)? Alias `walk_in_sales`.	

Questions

a) Q4 ÷ Q2 = _____ %. Compare with Day 3: GCash was **30.2%** of the *number* of sales. Is GCash's share of the *money* bigger or smaller? What does that suggest about GCash customers?

b) `COUNT(*)` gives 500 but `COUNT(customer_id)` gives 192. Why are they different? How does that connect to Q7?

Activity 5.2 — INNER JOIN, Two Tables (Pair, 20 pts)

Driver: _____ · Navigator: _____ (switch after Q9)

Q	Task	Rows / answer
Q8	Every sale with its **product name**: `sale_id, sale_date, product_name, quantity, total_amount`.	
Q9	Total revenue of the **Beverages** category. Alias `beverages_revenue`.	₱
Q10	Sale ID, **branch name** and total amount of every sale made in **Marikina**.	

Write your Q9 query here:

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c) In Q8, what error do you get if you write just `product_id` in the SELECT list instead of `s.product_id`? Why? *(Try it!)*

Activity 5.3 — The Joined Dataset, Three Tables (Pair, 10 pts)

Q11. Build one flat table with these 9 columns, **ordered by sale_id**:
`sale_id, sale_date, branch_name, product_name, category, quantity, unit_price, total_amount, payment_method`

Rows: _____ · First row's branch and product: _____

d) Why is this flat table easier for Excel or Power BI than the 4 separate tables?

Activity 5.4 — LEFT JOIN: What Is Missing? (Pair, 15 pts)

Q	Task	Answer
Q12	Which **products** have never been sold? `product_id, product_name`	

Q	Task	Answer
Q13	Which **branch** has no sales yet? `branch_id, branch_name, status`	
Q14	Which **suki customers** have never bought anything? `customer_id, first_name, last_name, joined_on`	

e) Look at the `joined_on` dates in Q14. What do these customers have in common? What could Aling Nena do for them?

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f) Change Q13's `LEFT JOIN` to `INNER JOIN` and run it. How many rows now? _____ Explain.

Activity 5.5 — Export to CSV, Import into Excel, Reconcile (Pair, 20 pts)

1. Run **Q11**. In the result panel click **EXPORT** → **Save results as CSV**. The *Save As* window opens in your lab folder: type the name `Day05_Joined_Sales.csv` → **Export**. *(It saves all 500 rows, even though the panel shows only 50 at a time.)*
2. Open a blank Excel workbook → **Data** → **From Text/CSV** → choose the file → **Import** (on some PCs the button says **Open**) → check *Delimiter: Comma* → **Load**.
3. Click an empty cell to the right of the table (e.g. **L2**) and type the checks below. Tip: type `=SUM(` and click the header of the *total_amount* column; Excel writes the table reference for you.
4. Save the workbook as `Day05_<Surname1>_<Surname2>.xlsx`.

Check	Excel result	SQL result	Match? ✓/ ✗
Rows `=ROWS(Day05_Joined_Sales[sale_id])`		Q1 =	
Revenue `=SUM(Day05_Joined_Sales[total_amount])`		Q2 =	
GCash revenue `=SUMIF(…[payment_method],"GCash",…[total_amount])`		Q4 =	
Beverages revenue `=SUMIF(…[category],"Beverages",…[total_amount])`		Q9 =	

g) Your partner opened the CSV by **double-clicking** instead. Name one problem that can happen, and why *From Text/CSV* is safer.

Pair Log (5 pts) *(Integrated: Lead small teams)*

Query / step	Driver	Navigator	One thing the navigator caught or suggested
Q8–Q9			
Q10–Q11			
Q12–Q14			
Export / import			

What did you do well as a team? What will you do differently next time?

Daily Reflection Log

1. Explain INNER JOIN vs LEFT JOIN to a friend using a class attendance list:

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2. The query I'm proudest of today, and why:

Submit: Day05_<Surname1>_<Surname2>.sql (Q1–Q14), Day05_Joined_Sales.csv, the reconciled Excel workbook, and this sheet with the pair log.